

SIMATS ENGINEERING ASSESSMENT TOOL 2

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA0702

COURSE OUTCOME COVERED

CO5: Measure network performance using key metrics with simulation tools.
(BL4,BL5)

Assessment Tool Weightage: 30%

Annexure A

Pseudocode Writing Task (Algorithm Design)

Code of Conduct:

I K.Yamini (1925720) Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

K.Yamini

Annexure A

Pseudocode Writing Task (Algorithm Design)

1. In a network simulation using Cisco Packet Tracer, a student records total data received as 8000 bits in 4 seconds and observes that 200 packets were sent while 180 packets were received. Write a pseudocode algorithm to calculate the network throughput in bits per second and the packet loss percentage based on the given values. The algorithm should include validation to ensure that time and total packets sent are not zero to avoid errors. It should display both the throughput and packet loss results clearly. Add logic to classify the network as Efficient if throughput is greater than 1500 bps and packet loss is less than 10%, otherwise classify it as Inefficient. Finally, ensure the algorithm outputs a complete summary of the network performance based on the calculated metrics.
2. An organization has multiple network links connecting its branches. The administrator wants to monitor bandwidth utilization and identify overloaded links. Write a pseudocode algorithm that periodically collects the bandwidth usage of each network link and stores the values in an array or list. Use a loop to continuously monitor all links. Add conditions to detect when bandwidth utilization exceeds 80% and recommend switching traffic to a backup link. Explain how this algorithm supports efficient bandwidth management and prevents network congestion.
3. A company has six branch offices connected through multiple network links. Each link has different bandwidth, delay, and utilization values, which change throughout the day. The network administrator wants to continuously identify the best communication path between the headquarters and each branch based on overall link quality rather than just the shortest distance. Write a pseudocode algorithm that periodically collects the bandwidth, delay, and utilization of every network link and

stores the values in suitable arrays or lists. Use a loop to repeat the monitoring process at regular intervals. Calculate a link quality score for each path using the collected metrics and select the path with the highest score. Include conditions to detect when a link becomes overloaded or fails, automatically selecting an alternate path and generating an alert for the administrator. Explain how your algorithm improves network reliability, routing efficiency, and fault tolerance while adapting to changing network conditions.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding	20%	Identifies all inputs, outputs, constraints accurately	Minor missing elements	Partial understanding	Incorrect interpretation
Algorithm Design	30%	Complete, correct, and logically sound solution	Minor logical gaps	Partially correct logic	Incorrect approach
Use of Control Structures	20%	Efficient and appropriate use of loops, conditions, functions/modules	Minor inefficiencies	Limited or basic usage	Incorrect or missing constructs
Pseudocode Structure	10%	Well-structured, clear, consistent format, easy to follow	Mostly clear with minor issues	Difficult to follow in parts	Poorly structured and unclear
Completeness	20%	Handles all cases;	Handles most cases;	Handles basic cases only	Incomplete or inefficient

CO5-TOOL-2Network Throughput and Packet Loss

Given:

- Total data received = 8000 bits
- Time = 4 seconds
- Packets sent = 200
- Packets received = 180

Formula:

- $\text{Throughput} = \text{Total data received} / \text{Time}$
- $\text{Packet Loss \%} = ((\text{Packets sent} - \text{Packets received}) / \text{Packets sent}) \times 100$

Pseudo code:

BEGIN

INPUT data, time, packets sent, packets received.

IF time = 0 OR packets sent = 0

DISPLAY "Invalid input"

ELSE

throughput = data / time

packet lost = packets sent - packets received

Loss = (lost packets / packets sent) * 100

DISPLAY throughput

DISPLAY Loss

IF throughput > 1500 AND Loss < 10

DISPLAY "Network is efficient"

ELSE

DISPLAY "Network is inefficient"

END IF

END IF

END

Result:-

Throughput :

$$\frac{8000}{4} = 2000 \text{ bps}$$

Packet Loss :

$$((200-186)/200) \times 100 = 10\%$$

- since throughput is greater than 1500 bps but packet loss is not less than 10%, the network is inefficient.

2. Bandwidth utilization monitoring

objective:

The algorithm continuously monitors the bandwidth utilization of multiple network links. if utilization exceeds 80%, it recommends moving traffic to a backup link.

Pseudocode:

BEGIN

INPUT number of Links

WHILE TRUE

FOR $i = 1$ TO number of Links

INPUT utilization $[i]$

```
IF utilization[i] > 80
```

```
    DISPLAY "Link", i, "is overload"
```

```
    DISPLAY "switch traffic to backup link"
```

```
ELSE
```

```
    DISPLAY "Link", i, "is normal"
```

```
END IF
```

```
END FOR
```

```
WAIT monitoringInterval
```

```
END WHILE
```

```
END
```

Explanation:-

- The number of network links is taken as input.
- An array stores the bandwidth utilization of each link.
- A WHILE LOOP continuously monitors the network.
- A FOR LOOP checks every link one by one.
- If utilization is greater than 80%, the link is considered overloaded.
- The algorithm recommends switching traffic to a backup link.
- otherwise, the link is considered normal.
- After checking all links, the algorithm waits for the next monitoring interval.
- continuous monitoring helps identify overloaded links early.
- Redirecting traffic helps reduce congestion and improve bandwidth management.

This matches the requirements for periodic collection, arrays loops, overload detection, and backup-link recommendation.

3. Best Communication Path Selection

Objective:

select the best network path based on bandwidth, delay, and utilization instead of simply selecting the shortest path. The algorithm must also handle overloaded or failed links.

Pseudocode:

BEGIN

INPUT number of Links

bestscore = -1

best path = -1

WHILE TRUE

FOR i = 1 to number of Links

INPUT bandwidth, delay, utilization, status

IF status = "FAILED" OR utilization > 80

DISPLAY "Link unavailable"

CONTINUE

END IF

score = (bandwidth / (delay + 1)) * (100 - utilization)

IF score > bestscore

bestscore = score

best path = i

END IF

END FOR

IF best path = -1

```
    DISPLAY "No suitable path available"  
ELSE  
    DISPLAY "Best path = ", best path  
    DISPLAY "Best score = ", best score  
END IF  
WAIT monitoringInterval  
END WHILE  
END
```

Explanation :

- Bandwidth : Higher bandwidth improves the quality score.
- Delay : Lower delay gives a better score.
- Utilization : Lower utilization gives a better score.
- Failed links : They are ignored.
- overloaded links : Links with utilization above 80% are avoided.
- The algorithm compares all available links and selects the one with the highest quality score.
- If the selected link fails or becomes overloaded during the next monitoring cycle, another suitable link can automatically become the best path.
- If no suitable link exists, an alert is generated for the administrator.

Therefore, the algorithm improves network reliability, routing efficiency, fault tolerance, and adaptability to changing network conditions as required by the question.